

BlueStock MF Capstone Project

Mutual Fund Analytics Platform

End-to-End Data Engineering, ETL Pipeline & Interactive Dashboard

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1. Executive Summary

BlueStock MF Capstone is an end-to-end mutual fund analytics platform that analyzes 40 equity schemes across 10+ Asset Management Companies in India. The project covers data ingestion from AMFI India, ETL processing, SQLite database storage, EDA visualizations, performance metrics computation, interactive Power BI dashboard, and advanced analytics including VaR, cohort analysis, and a fund recommender system.

Key Highlights

40 Schemes	87K+ Rows	₹81L Cr AUM	6 Deliverables
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2. Data Sources

Dataset	Description	Records
fund_master.csv	40 schemes — AMFI codes, AMC, expense ratio	40 rows
nav_history.csv	Daily NAV Jan 2022 to May 2026	46,000 rows
aum_by_fund_house.csv	Quarterly AUM for 10 fund houses	90 rows
monthly_sip_inflows.csv	AMFI Monthly SIP data	48 rows
investor_transactions.csv	SIP/Lumpsum/Redemption — 5,000 investors	32,778 rows
benchmark_indices.csv	NIFTY50, NIFTY100, BSE SmallCap	8,050 rows

3. ETL Pipeline

- Data ingestion from 10 CSV files using Pandas
- Data cleaning: date parsing, missing value handling, duplicate removal
- Forward-fill for NAV holidays and weekends
- Star schema design in SQLite with 6 tables: dim_fund, fact_nav, fact_transactions, fact_performance, fact_aum, fact_benchmark
- Live NAV fetch from mfapi.in REST API for 5 key schemes

4. EDA Findings

Metric	Value
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Total Industry AUM	₹81 Lakh Crore
SIP Inflows (Dec 2025)	₹31,002 Crore — All-time High
Folio Count Growth	13.26 Cr (Jan 2022) to 26.12 Cr (Dec 2025)
Top State by SIP	Punjab, Tamil Nadu, Madhya Pradesh
Top AMC by AUM	SBI MF — ₹12.5 Lakh Crore
Gender Split	Male 66.5% Female 33.5%
T30 vs B30	T30 cities: 66.3% of transactions

5. Performance Metrics

5.1 Top Funds by Sharpe Ratio

Fund Name	Sharpe Ratio	1yr Return
Mirae Asset Large Cap - Regular	1.45	15.12%
Kotak Flexicap - Regular	1.31	17.35%
Mirae Asset Tax Saver - Regular	1.23	14.78%
SBI Bluechip - Regular	1.21	12.42%
ICICI Pru Midcap - Regular	1.18	14.02%

5.2 Top Funds by Alpha

Fund Name	Alpha	Beta
SBI Small Cap - Regular	+30.34%	-0.02
DSP Small Cap - Regular	+30.06%	0.01
ICICI Pru Midcap - Regular	+29.26%	0.00
Mirae Asset Tax Saver - Regular	+28.27%	0.02
Kotak Flexicap - Regular	+27.33%	-0.02

6. Power BI Dashboard

The interactive Power BI dashboard consists of 4 pages, each with slicers for Fund House, Category, Plan, State, and Date Range.

6.1 Page 1 — Industry Overview

KPI cards showing Total AUM (₹81 Lakh Crore), SIP Inflows, Total Folios, and Total Schemes. Includes AUM Growth Trend line chart and AUM by Fund House bar chart.

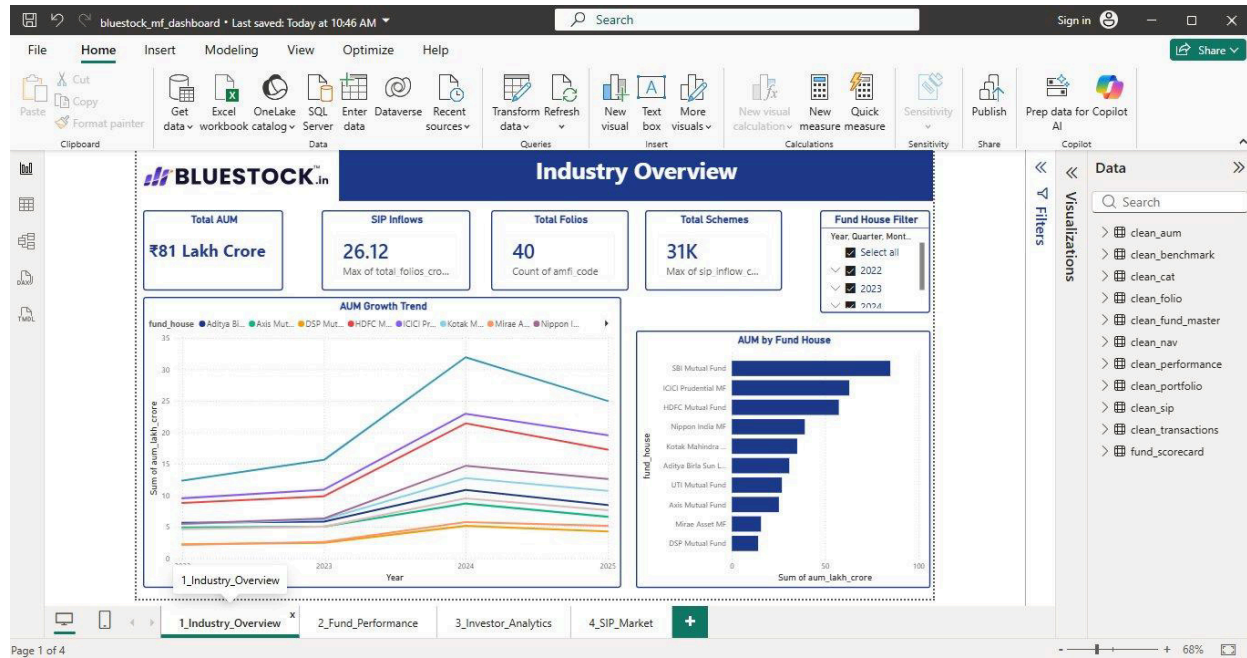


Figure 1: Industry Overview — KPI Cards, AUM Growth Trend & AUM by Fund House

6.2 Page 2 — Fund Performance

NAV Line Chart vs Benchmark, Risk vs Return Scatter Plot, and Fund Scorecard table with slicers for category, plan, and fund house.

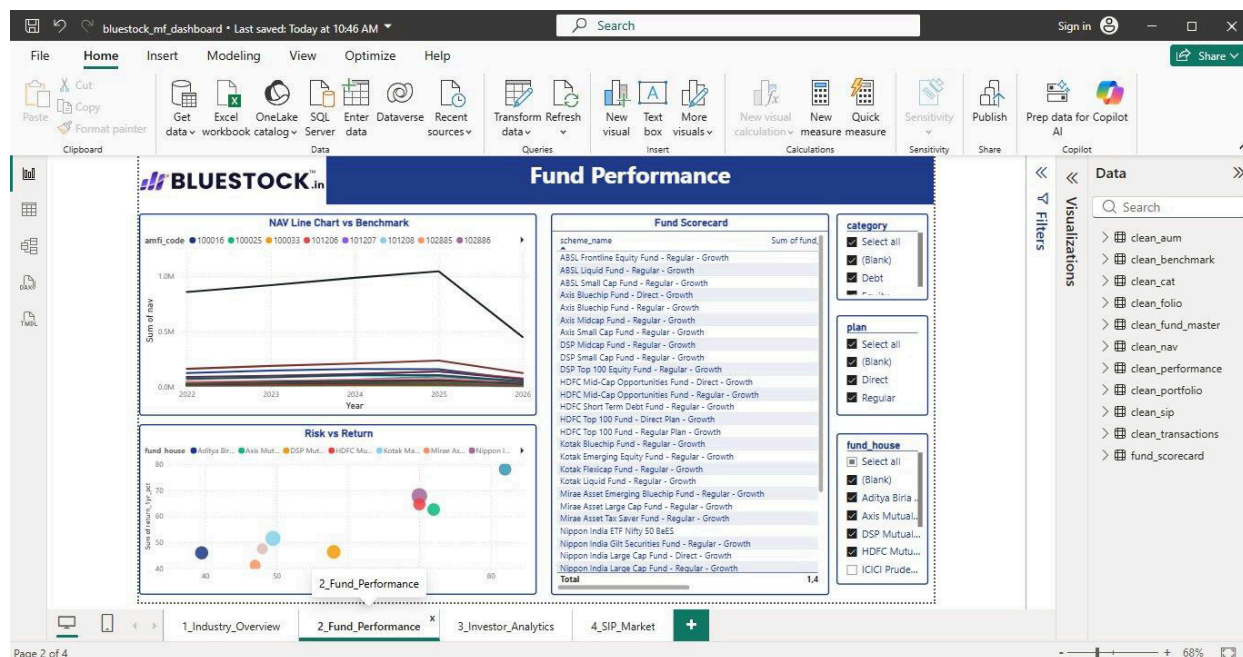


Figure 2: Fund Performance — NAV Benchmark, Risk-Return Scatter & Fund Scorecard

6.3 Page 3 — Investor Analytics

Transaction Types donut chart, Amount by State stacked bar chart, with slicers for city tier, age group, and state.

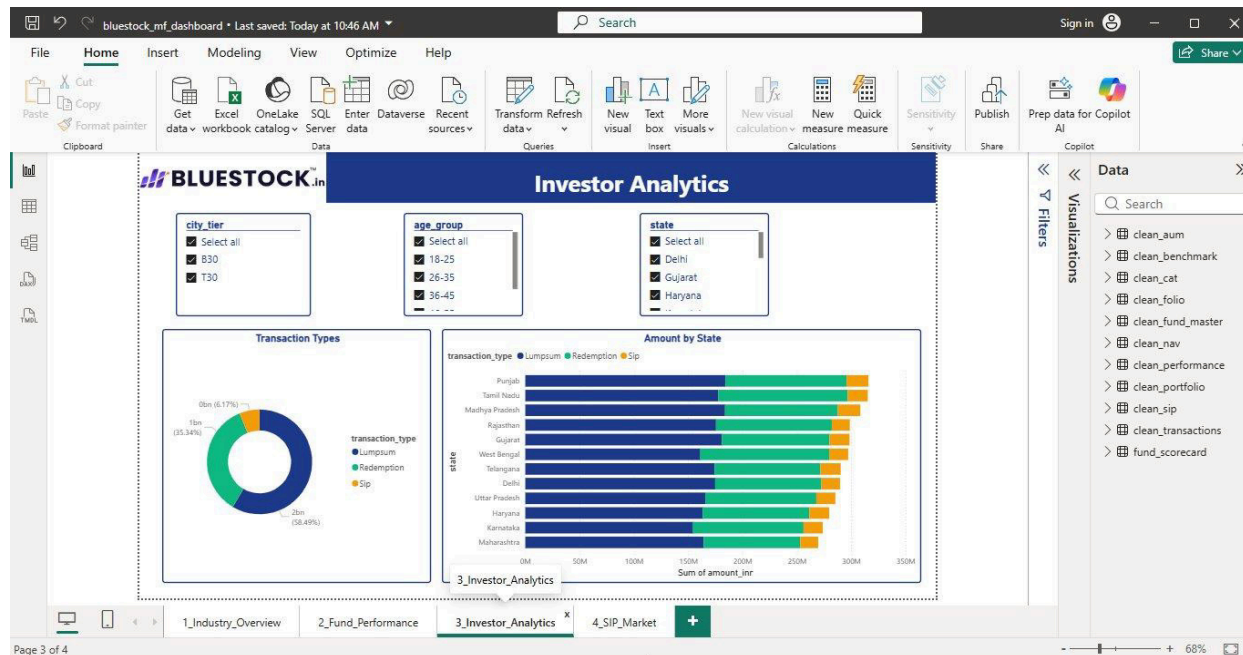


Figure 3: Investor Analytics — Transaction Types, State Distribution & Slicers

6.4 Page 4 — SIP & Market Trends

SIP + Nifty50 dual-axis chart, Category Inflow Heatmap matrix, and Top 5 Categories bar chart with date and category slicers.

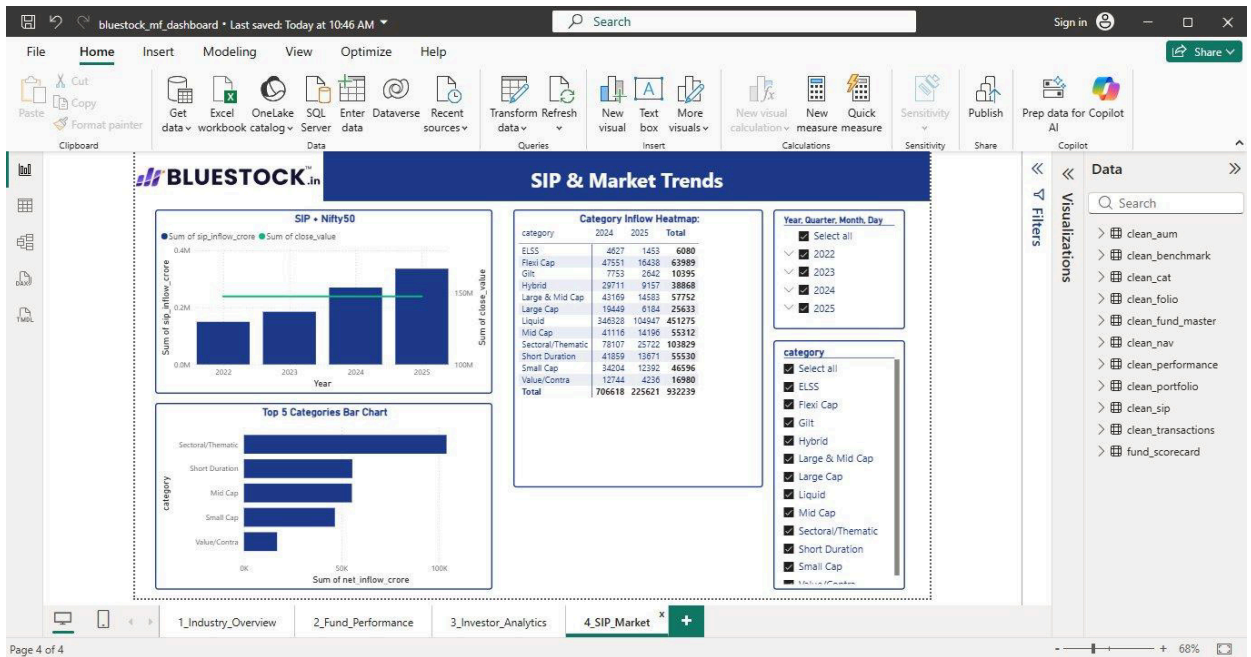


Figure 4: SIP & Market Trends — Dual-axis Chart, Category Heatmap & Top 5

7. Advanced Analytics

7.1 Value at Risk (VaR & CVaR)

- Small Cap funds show highest VaR at -2.7% per day (95% confidence)
- Liquid funds: lowest risk with VaR near -0.1% per day
- CVaR confirms Small Cap tail risk is significantly higher

7.2 Investor Cohort Analysis

- 2024 cohort: 4,624 investors, average SIP amount ₹10,997
- 2025 cohort: 138 investors, average SIP amount ₹13,505
- Newer investors commit higher SIP amounts per transaction

7.3 SIP Continuity Analysis

- 1,332 out of 1,362 investors (97.8%) flagged as at-risk
- At-risk criterion: average gap between SIP transactions > 35 days
- Indicates high SIP dropout risk — key business concern

7.4 Fund Recommender System

Risk Profile	Recommended Funds (Top 3 by Sharpe)
Low	ICICI Pru Liquid (7.68) • Kotak Liquid (6.18) • ABSL Liquid (5.14)

Moderate	Mirae Asset Large Cap (1.06) • HDFC Top 100 (1.06) • ICICI Pru Bluechip (1.03)
High	Kotak Emerging Equity (0.96) • ICICI Pru Midcap (0.95) • SBI Small Cap (0.94)

8. Limitations

- Only 40 schemes analyzed — not all AMCs covered
- Live data limited to 6 schemes via mfapi.in API
- Dashboard not published to Power BI Service
- No machine learning models deployed
- Investor transaction data is simulated, not real

9. Conclusion

This capstone project demonstrates end-to-end data engineering and analytics skills using Python, SQL, and Power BI. The pipeline processes raw CSV data through cleaning and transformation into an interactive dashboard providing actionable insights for mutual fund investors.

GitHub: https://github.com/Vankireddysomeswarreddy/bluestock_mf_capstone

10. Project Deliverables

Day	Task	Output Files
Day 1	Project Setup + ETL Pipeline	data_ingestion.py, live_nav_fetch.py, requirements.txt
Day 2	Data Cleaning + SQLite DB	10 cleaned CSVs, bluestock_mf.db, schema.sql, queries.sql
Day 3	EDA Analysis — 15+ Charts	03_eda_analysis.ipynb, 15 PNG chart exports
Day 4	Performance Analytics	fund_scorecard.csv, alpha_beta.csv, benchmark chart
Day 5	Power BI Dashboard (4 pages)	bluestock_mf_dashboard.pbix, Dashboard.pdf
Day 6	Advanced Analytics + Risk	var_cvar_report.csv, recommender.py, sector_hhi.csv
Day 7	Final Report + Presentation	README.md, Final_Report.docx, Presentation.pptx

Submitted by: Vankireddy Someswarreddy | BlueStock Fintech Internship | july 2026